STPM Physics · Semester 3 · Chapter 26

NUCLEAR PHYSICS

26.1 Nucleus

DEFINITIONS & LAWS

- The **proton (atomic) number** Z is the number of protons in the nucleus; the **nucleon (mass) number** A is the total number of protons and neutrons; the number of neutrons N = A − Z.
- **Isotopes** are nuclides of the same element (same Z) with different numbers of neutrons (different A).
- A **nuclide** is a particular species of nucleus with a specified proton number Z and nucleon number A.
- The **unified atomic mass unit** (u) is one twelfth of the mass of a neutral carbon-12 atom; 1 u = 1.66×10^{-27} kg ≡ 931.5 MeV.
- The **mass defect** is the difference between the total mass of the separate nucleons and the mass of the nucleus.
- The **binding energy** is the energy required to separate a nucleus into its individual nucleons; $B = \Delta mc^2$.
- The **binding energy per nucleon** is the binding energy of a nucleus divided by its number of nucleons. It is a measure of nuclear stability: a nucleus with a greater binding energy per nucleon is more stable; the curve rises steeply for light nuclei, peaks at about 8.8 MeV near ^{56}Fe (A ≈ 56) and falls slowly for heavy nuclei.
- The nuclear radius $R = R_0A^{1/3}$, so the nuclear density is the same for all nuclei (about 2×10^{17} kg m^{-3}).

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Mass defect	$\Delta m = Zm_p + (A - Z)m_n - m_{nucleus}$	Δm = mass defect (u); Z = proton number; A = nucleon number; m_p, m_n = masses of proton and neutron (u); $m_{nucleus}$ = mass of nucleus (u); use atomic masses with m_H when electrons are included
Binding energy	$B = \Delta mc^2$; binding energy per nucleon = $\frac{B}{A}$	B = binding energy; Δm = mass defect (u); c = speed of light; 1 u = 931.5 MeV; A = nucleon number
Nuclear radius	$R = R_0A^{1/3}$	R = nuclear radius (m); $R_0 \approx 1.2\times10^{-15}$ m (1.2 fm); A = nucleon number
Nuclear density	$\rho = Am_u/(\frac{4}{3}\pi R^3) = \frac{3m_u}{4\pi {R_0}^3}$	ρ = density (kg m^{-3}); m_u = atomic mass unit (kg); R = nuclear radius (m); $R_0 = 1.2\times10^{-15}$ m; independent of A

- 1 MeV = 1.60×10^{-13} J; c = 3.00×10^8 m s^{-1}.

TIPS TO ANSWER

- For a nuclide, state the proton, neutron and nucleon numbers from the symbol $^{A}_{Z}X$; write $N = A - Z$ explicitly.
- For binding energy: $\Delta m \rightarrow B = \Delta mc^2$ (or $\times$ 931.5 MeV u^{-1}) $\rightarrow \frac{B}{A}$; show masses to enough decimal places.
- Use $R \propto A^{1/3}$ ratios; show the cube root clearly and state the ratio to 2 or 3 s.f.
- For the binding-energy curve: label axes ($\frac{B}{A}$ against A), mark the peak near $A \approx 56$, and state that fusion of light nuclei and fission of heavy nuclei both release energy because $\frac{B}{A}$ increases.

SECTION A TRAPS

Trap: Taking neutron number as A instead of A − Z (e.g. 64 neutrons for $^{64}_{29}Cu$).

Trap: Quoting total binding energy when the question asks for binding energy per nucleon; divide by A.

Trap: Using $R \propto A$ instead of $A^{1/3}$; doubling A increases R by 1.26, not 2.

Trap: Placing the peak of the $\frac{B}{A}$ curve at helium or uranium; it is at iron ($A \approx 56$).

TRICKS

Trick: Cube-root shortcuts: 8 → 2, 27 → 3, 64 → 4, 125 → 5, 216 → 6.

Trick: Density of the nucleus is independent of A: any option that depends on A is wrong.

26.2 Radioactivity

DEFINITIONS & LAWS

- **Radioactive decay** is the spontaneous and random disintegration of an unstable nucleus with the emission of α, β or γ radiation.
- Radioactive decay is **random** and **spontaneous**. Random: it is impossible to predict when a particular nucleus will decay, and every nucleus has the same probability of decay per unit time. Spontaneous: the decay is not affected by external physical conditions such as temperature and pressure.

- The **activity** A is the rate of decay of the nuclei, $A = -\frac{dN}{dt} = \lambda N$, measured in becquerel (Bq; 1 Bq = 1 decay per second).
- The **decay constant** λ is the probability of decay per unit time of a nucleus.
- The **half-life** $t_{½}$ is the time taken for the number of undecayed nuclei (or the activity) to fall to half of its initial value.
- α emission reduces A by 4 and Z by 2; β^- emission raises Z by 1 with A unchanged; γ emission changes neither A nor Z.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Decay law	$N = N_0e^{-\lambda t}$; $A = A_0e^{-\lambda t}$	N = number of undecayed nuclei; A = activity (Bq); N_0, A_0 = values at t = 0; λ = decay constant (s^{-1}); t = time (s)
Half-life	$t_{½} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$	$t_{½}$ = half-life (s); λ = decay constant (s^{-1})
Half-life form of decay law	$N = N_0(½)^{t/t_{½}}$; fraction remaining = $(½)^{t/t_{½}}$	N = number of undecayed nuclei; N_0 = initial number; t = time (s); $t_{½}$ = half-life (s)
Activity	$A = \lambda N$; $A \propto N$	A = activity (Bq); λ = decay constant (s^{-1}); N = number of undecayed nuclei
Linearised decay law	$\ln A = \ln A_0 - \lambda t$	A = activity (Bq); A_0 = initial activity (Bq); λ = decay constant (s^{-1}); t = time (s); graph of ln A against t is a straight line with gradient −λ

TIPS TO ANSWER

- Convert the half-life to seconds when the activity is required in Bq: $t_{½}$ in s → $\lambda = \frac{\ln 2}{t_{½}}$ → $A = \lambda N$.
- State the number of half-lives $n = \frac{t}{t_{½}}$ first, then fraction = $(½)^n$; non-integer n is allowed with the exponential form.
- Show the substitution, the evaluated result to 3 s.f., and the unit (Bq, s^{-1} or mass in the unit asked).
- For explanation questions, state 'random and spontaneous' and refer to constant probability of decay per unit time.

SECTION A TRAPS

Trap: Taking fraction remaining after n half-lives as $\frac{1}{2n}$ or $1 - \frac{n}{2}$; the correct value is $(½)^n$.

Trap: Leaving $t_{½}$ in days or years when computing A in Bq; λ must be in s^{-1}.

Trap: Equating half-life with the time for all nuclei to decay, or half of the 'total life'. The half-life applies at any starting point.

Trap: Confusing β^- with α: β^- increases Z by 1 and leaves A unchanged.

TRICKS

Trick: Count half-lives: 28 days with $t_{½}$ = 8.0 days is 3.5 half-lives, so fraction = $(½)^{3.5} \approx 0.088$.

Trick: The mass of undecayed material and the activity fall by the same fraction as N.

26.3 Nuclear Reaction

DEFINITIONS & LAWS

- In a nuclear reaction, mass number, charge (atomic number), energy and momentum are conserved; rest mass is not conserved.
- The **Q-value** of a reaction is the energy released (Q > 0) or absorbed (Q < 0), Q = (total mass of reactants − total mass of products)c^2.
- **Nuclear fission** is the splitting of a heavy nucleus (e.g. ^{235}U, after absorbing a neutron) into two lighter nuclei with the release of energy and neutrons; each fission of ^{235}U releases about 200 MeV.
- **Nuclear fusion** is the joining of two light nuclei to form a heavier nucleus with the release of energy; it requires a very high temperature to overcome electrostatic repulsion.
- Energy is released in fission and fusion because the products have a greater binding energy per nucleon, so their total mass is less than that of the reactants.
- A **chain reaction** is sustained when, on average, at least one of the secondary neutrons from each fission causes another fission.
- A **thermal neutron** is a slow neutron whose kinetic energy is about the same as the thermal energy of particles at room temperature (about 0.025 eV); thermal neutrons are the most likely to cause fission of ^{235}U.
- A **thermonuclear reaction** is a fusion reaction that occurs when the temperature of the light nuclei is raised to about 10^8 K, so that they collide with enough kinetic energy to overcome the electrostatic repulsion and fuse.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Q-value of a reaction	$Q = \Delta mc^2 = (\Sigma m_{reactants} - \Sigma m_{products}) \times 931.5$ MeV	Q = energy released (MeV); Δm = mass difference (u); masses in u; c = speed of light

Quantity / law	Formula	Symbols, units & conditions
Mass-energy equivalence	$E = \Delta mc^2$	E = energy (J); Δm = mass difference (kg); c = 3.00×10^{8} m s^{-1}; 1 u = 1.66×10^{-27} kg
Nuclear equation	ΣA and ΣZ equal on both sides; e.g. $^{2}H + {}^{3}H \rightarrow {}^{4}He + n$	A = nucleon number; Z = proton number; both conserved in a nuclear reaction
Power from fission	P = (fission rate) × (energy per fission)	P = power (W); fission rate in s^{-1}; energy per fission in J
Mass of ^{235}U used	$\text{Mass} = \frac{\text{number of fissions} \times 235\ \text{g}}{N_A}$	Mass in g; N_A = Avogadro constant (mol^{-1})

- 1 eV = 1.60×10^{-19} J; $N_A = 6.02\times10^{23}$ mol^{-1}.

TIPS TO ANSWER

- Balance the nuclear equation first, checking the top (A) and bottom (Z) numbers separately.
- Calculate the mass defect with masses to 6 decimal places, then multiply by 931.5 MeV u^{-1}; give Q to 3 s.f. with unit, and state the sign.
- For power problems, convert MeV to J, then find the fission rate, then the amount of fuel consumed in the stated time (seconds in a day = 86 400).
- For a decay such as α emission, use the masses of the nuclei given; do not add electrons unless atomic masses are specified.

SECTION A TRAPS

Trap: Forgetting to convert Δm to energy (leaving the answer in u) or multiplying by 931.5 when Δm is already in kg.

Trap: Subtracting reactants from products, giving a negative Q for an exothermic reaction.

Trap: Using the mass of the electron in nuclear masses inconsistently: atomic and nuclear masses must not be mixed.

Trap: Taking a molar mass of 235 kg mol^{-1} instead of 0.235 kg mol^{-1} (235 g mol^{-1}).

TRICKS

Trick: If the reaction releases energy, total mass decreases: Δm = (before) − (after) must be positive.

> **Trick:** Energy check: 1 u ≈ 931.5 MeV, so Δm of about 0.02 u gives about 19 MeV: any option far from that order is wrong.